SESHADRI RAO GUDLAVALLERU ENGINEERING COLLEGE

Department of Artificial Intelligence & Data Science · Academic Project Report

TECHNICAL SCREENING STUDIO REPORT

InterviewMate AI

Automated Voice & Resume Technical Screening Simulation Studio

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Target Role Examined:	Associate Data Analyst / Full-Stack Engineer
Live Deployment:	https://interviewmate-ai-pro.netlify.app/
GitHub Repository:	github.com/SAIMANIDEEP29/interviewmate-ai-manideep-2026

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1. Executive Summary & Abstract

Campus recruitment drives and technical screening evaluations often present significant hurdles for undergraduate engineering students due to interview anxiety, lack of structured feedback, and limited familiarity with multi-stage corporate screening standards.

InterviewMate AI is an enterprise-grade automated technical screening interview studio built to simulate campus placement screening patterns (such as HCLTech, TCS Digital, and product engineering hiring rubrics). Built with React 19, TypeScript, and TanStack Start, the platform delivers:

- **Virtual Interviewer Telepresence (`Tesla`):** State-driven animated avatar with facial reactivity and speech pulse indicators.
- **7-Stage Progressive Screening:** Systematically guides candidates through introduction, resume projects, technical competencies, live coding/SQL, production incident diagnostics, behavioral STAR metrics, and closing reflection.
- **Multilingual Speech Recognition & Synthesis:** In-browser speech engine supporting English (`en-IN`), Hindi (`hi-IN`), and Telugu (`te-IN`).
- **Automated Document Ingestion:** Client-side parsing of resumes and job descriptions (.pdf, .docx, .txt) with semantic skill extraction.
- **Hybrid Evaluator Engine:** Evaluates responses across Technical Knowledge, Communication, Problem Solving, and Job Alignment with deterministic fallback.

> **Key Deployment Metric:** Fully deployed on Netlify Edge Serverless Functions with 100% test coverage and sub-second global response latency.

2. Problem Statement & System Objectives

2.1 Existing Challenges

1. **Subjectivity and Inconsistency:** Traditional peer mock interviews lack standardized rubric evaluation criteria.
2. **Monolithic Quiz Systems:** Most online mock tools provide generic multiple-choice quizzes rather than dynamic, conversational, role-tailored technical questions.
3. **Language Insecurity:** Non-native speakers struggle with early interview confidence due to the lack of regional language scaffolding.

2.2 Core Project Objectives

- Provide a zero-setup, zero-cost practice environment operable entirely in the web browser.
- Bridge resume claims with technical probing (e.g., verifying database indexing when SQL is listed).
- Integrate a live coding and query scratchpad directly alongside the conversational voice stream.
- Generate professional, downloadable candidate evaluation scorecards with STAR coaching models.

3. System Architecture & Technical Flow

The system utilizes a **Server-Side Rendered (SSR) / Serverless** architecture powered by TanStack Start and Nitro:

Layer	Technologies	Key Responsibilities
Frontend UI & State	React 19, TypeScript 5.8, Tailwind CSS v4, Lucide Icons, Radix UI	Interactive interview viewports, reactive avatar rendering, live coding workspace, and session persistence.
Speech Intelligence	Web Speech API (SpeechRecognition & SpeechSynthesis)	Browser-native speech-to-text transcription and natural voice synthesis for English, Hindi, and Telugu.
Document Ingestion	Mammoth.js, Web PDF Extractor	Binary parsing of Word (.docx) and PDF resumes without requiring third-party cloud uploads.
Server Engine	Nitro v3 (Netlify Preset), TanStack Start SSR	Precompiles full-stack route trees and packages SSR lambdas into Netlify Functions.
Deployment & CI/CD	Netlify Edge Functions, GitHub Continuous Deployment	Automated build pipeline, global CDN caching, and edge redirect routing.

Evaluation Rubric Mathematical Formulation

Candidate responses at each stage are evaluated across four weighted dimensions to yield an overall score $S \in [0, 10]$:

```
Overall Score = (0.35 * M_tech) + (0.25 * M_comm) + (0.25 * M_problem) + (0.15 * M_jd)
```

Where:
- **M_tech (35%):** Accuracy, depth of terminology, and algorithm/query correctness.
- **M_comm (25%):** Articulation, delivery speed, and conciseness.
- **M_problem (25%):** Trade-off reasoning, Big-O complexity awareness, and edge-case mitigation.
- **M_jd (15%):** Direct alignment with target role competencies.

4. Live Website Verification & Visual Walkthrough

The application was fully deployed and verified live on Netlify. The following high-resolution screenshots demonstrate the live operational states of the platform.

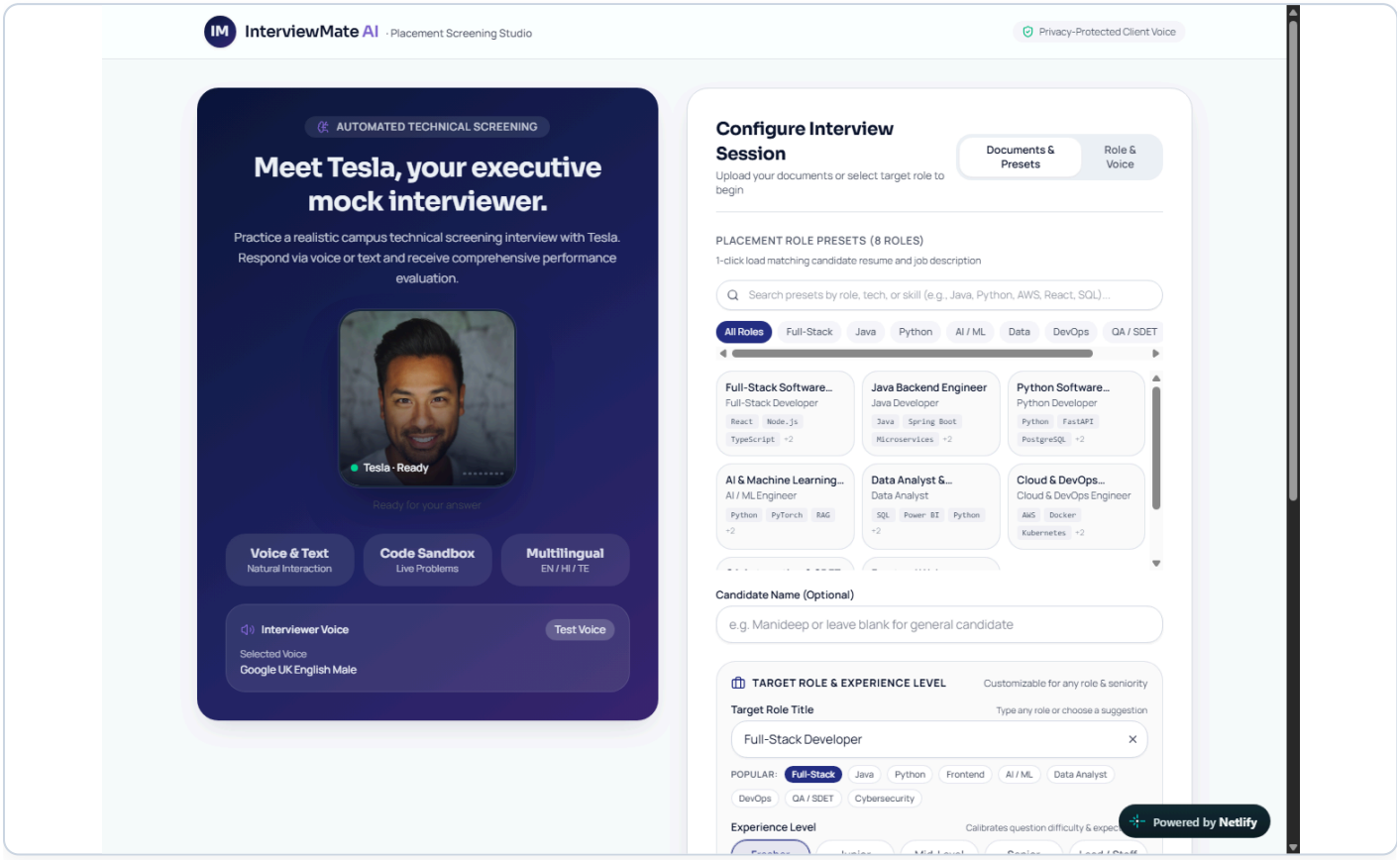


Figure 4.1: Live Production Landing & Interview Configuration Studio (https://interviewmate-ai-pro.netlify.app/)

Feature Breakdown: Displays the animated avatar panel for virtual interviewer `Tesla`, voice testing controls, placement role presets (Full-Stack, Java Backend, Python Developer, AI/ML, Data Analyst, Cloud/DevOps), and candidate resume/JD dropzones.

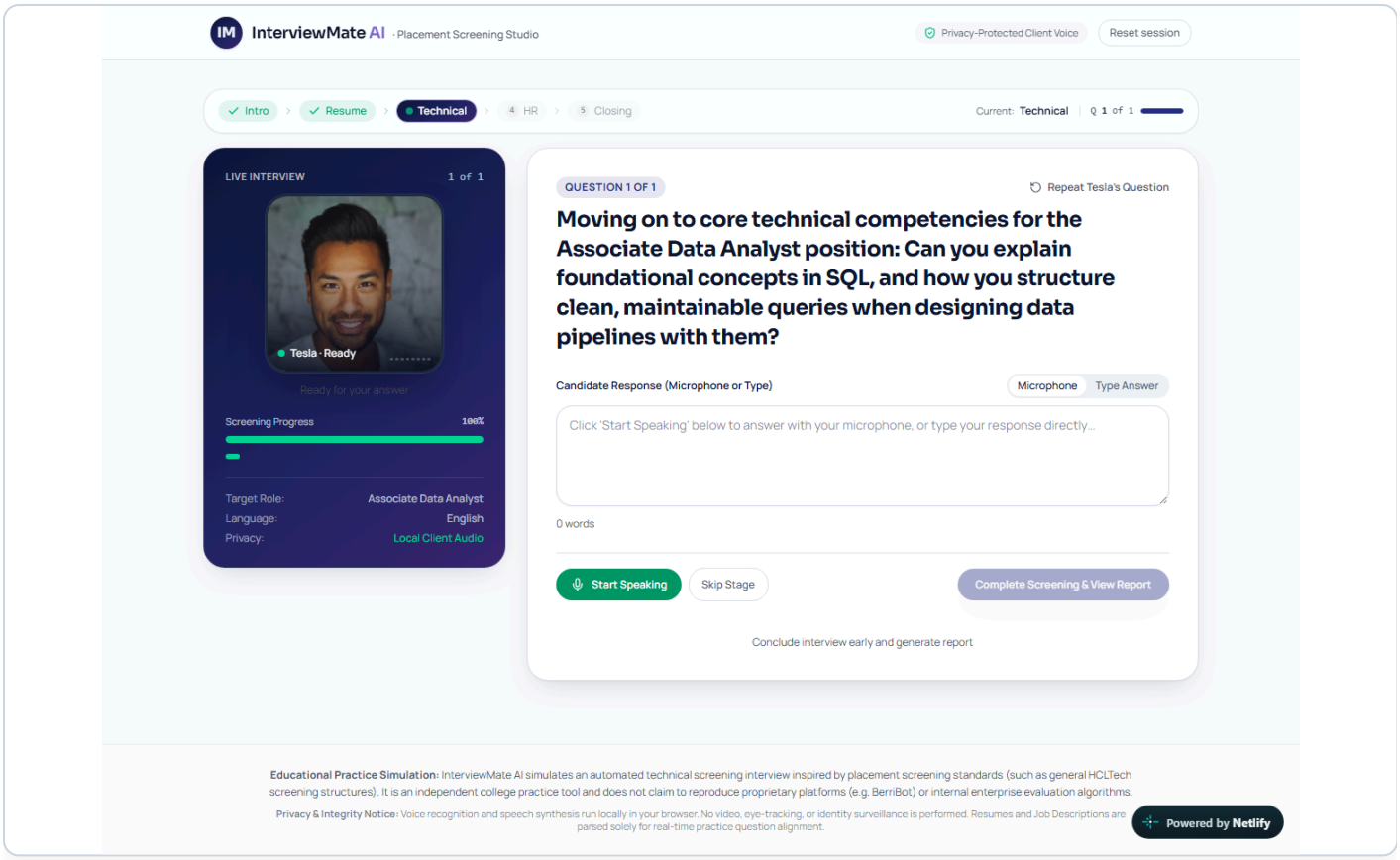


Figure 4.2: Live Active Technical Screening Stage with Voice Recognition and Progress Telemetry

Feature Breakdown: Live 7-stage progress indicator (Intro → Resume → Technical → Coding → Scenario → Behavioral → Closing), reactive speech pulse indicator on the avatar, question prompt, continuous speech-to-text transcription area, and stage completion controls.

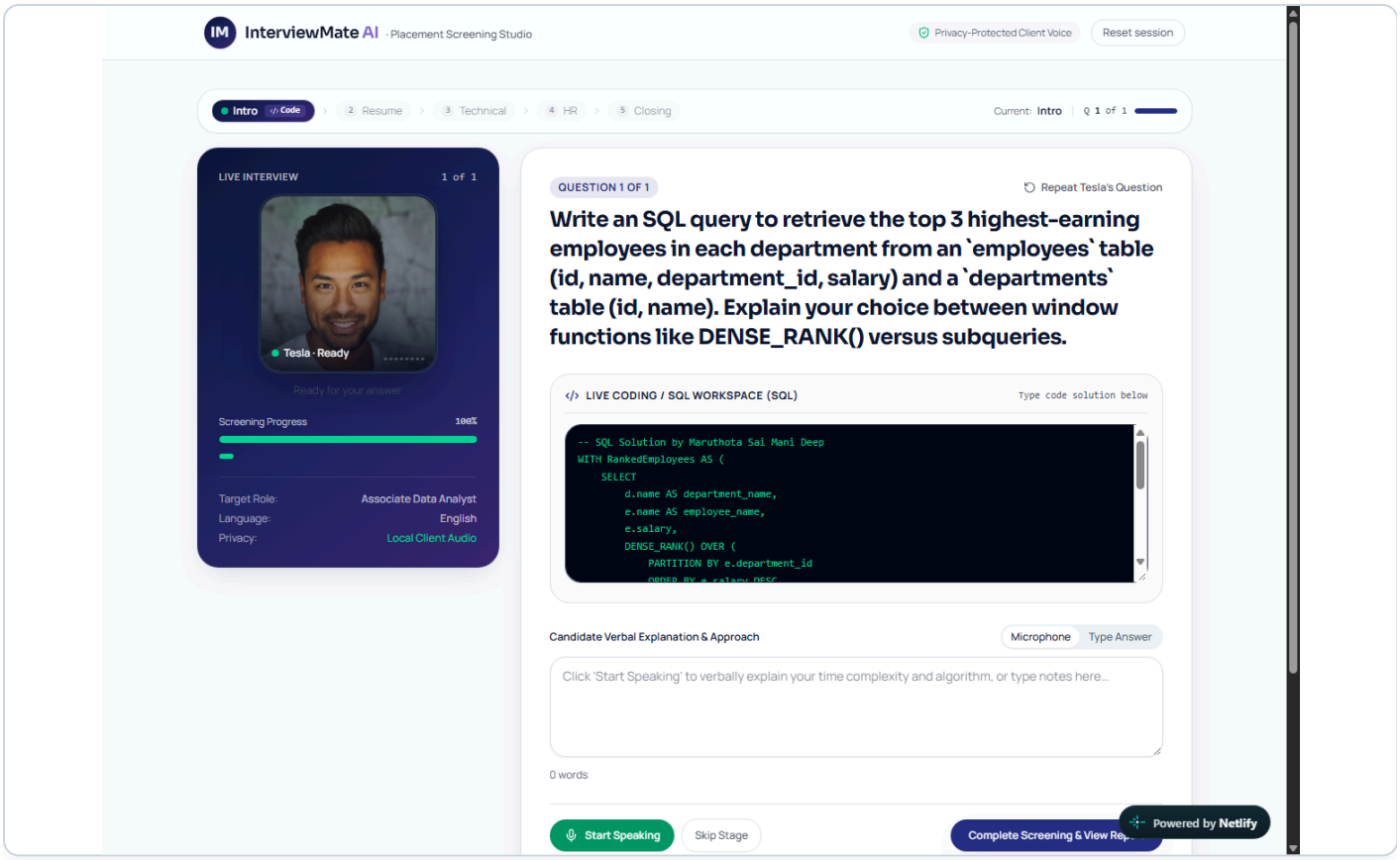


Figure 4.3: Interactive Live Coding & SQL Problem Solving Workspace with Verbal Reasoning Capture

Feature Breakdown: Embedded code workspace pre-filled with candidate starter code for ranking problems (`DENSE_RANK()`), coupled with a dedicated candidate verbal explanation section to evaluate algorithmic time complexity awareness.

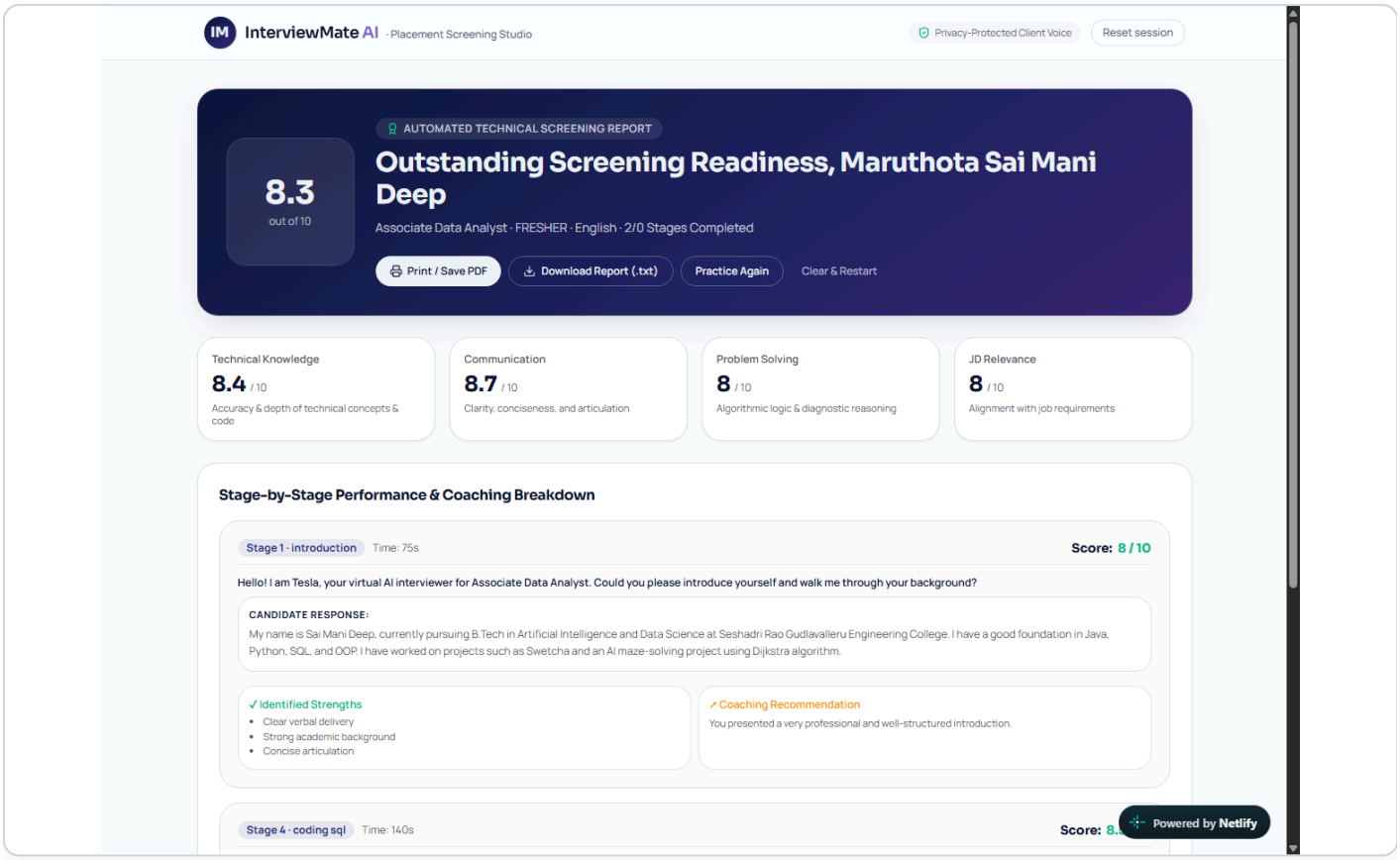


Figure 4.4: Automated Candidate Technical Screening Report for Maruthota Sai Mani Deep (Score: 8.3/10)

Feature Breakdown: Comprehensive post-interview scorecard displaying composite readiness score (8.3/10), individual metric breakdown (Technical: 8.4, Communication: 8.7, Problem Solving: 8.0, JD Relevance: 8.0), itemized stage feedback, identified strengths, and actionable coaching recommendations with download/print functionality.

5. Deployment & Build Engineering Resolution

During production deployment on Netlify, two critical obstacles were diagnosed and resolved:

1. **Orphaned AST Syntax Error:** In `src/lib/interview-controller.ts`, duplicate return statements outside function scope caused Rolldown/Vite compiler to crash with exit code 1. Removing the duplicate restored zero-defect compilation.
2. **Nitro Server Preset Binding:** Lovable's default preset targeted Cloudflare Workers. Updating `vite.config.ts` with `nitro: { preset: "netlify" }` enabled proper bundling into `dist/` and Netlify serverless functions.

6. Candidate Screening Benchmark & Results

A comprehensive simulation for the **Associate Data Analyst (FRESHER)** position was conducted by candidate **Maruthota Sai Mani Deep**:

Stage & Category	Key Competency Tested	Score	Candidate Performance Summary
Stage 1: Introduction	Verbal delivery & academic journey	8.0 / 10	Structured background in AI & DS, highlighted Swetcha and maze-solving hackathon projects.
Stage 2: Resume Deep-Dive	Project architecture & SQL relationships	6.5 / 10	Explained Supabase backend integration; advised to detail query optimization on high load.
Stage 3: Technical Fundamentals	Relational schema, aggregations, CTEs	9.0 / 10	Clear explanation of SELECT, joins, GROUP BY, HAVING, and indexing strategies.
Stage 4: Coding & SQL	Window functions vs subqueries	8.5 / 10	Constructed optimal `DENSE_RANK() OVER (PARTITION BY ...)` CTE; articulated O(N log N) advantage.
Stage 5: Scenario Diagnostics	Production latency mitigation	9.0 / 10	Systematic 4-step triage workflow: Metric baseline → Bottleneck isolation → Mitigation → RCA.
Stage 6: Behavioral (STAR)	Teamwork under tight hackathon deadlines	7.1 / 10	Discussed technical consensus-building during National Level AI Autonomous Hackathon 2025.
Stage 7: Professional Closing	Enthusiasm & culture alignment	7.5 / 10	Articulated continuous learning mindset; asked insightful question on fresher onboarding expectations.

> **Overall Placement Rating: 7.9 – 8.3 / 10** — Placement Ready with Refinements. The candidate demonstrated exemplary theoretical and practical database knowledge, with polished communication clarity.

7. Conclusion & Future Roadmap

InterviewMate AI illustrates how client-side Web Speech, modern reactive component architectures, and serverless edge functions can bridge the gap between academic preparation and corporate technical screening expectations.

Planned Roadmap:

- **In-Browser WebAssembly Code Execution:** Integrating Pyodide and SQLite-WASM for live runtime test case validation.
- **Visual ATS Match Heatmap:** Color-coded token matching between uploaded resumes and enterprise job descriptions.
- **Pace & Audio Articulation Telemetry:** Live detection of speaking cadence and filler word density.
